Smooth six-functor formalisms

Workshop “Alpbach 2026”

Gabriel Ribeiro (joint work with Martin Gallauer)

June 19, 2026

ETH Zürich

One reason why you should care

Counting rational points

Let X be a d-dimensional variety over a finite field $k = \mathbb{F}_q$. Given that you are here, you might care about $|X(k)|$. These numbers are related to deep results in arithmetic geometry. Notably, the Weil conjectures.

Weil famously suggested that his conjectures might be related to some imaginary cohomology theory, that later became known as ℓ-adic cohomology:

$$|X(k)| = \sum_{i=0}^{2d} (-1)^i \operatorname{tr}\left(\mathrm{Frob}_k \mid \mathrm{H}^i_c(X_{\bar{k}}, \mathbb{Q}_\ell)\right).$$

Counting rational points

Let X be a d-dimensional variety over a finite field $k = \mathbb{F}_q$. Given that you are here, you might care about $|X(k)|$. These numbers are related to deep results in arithmetic geometry. Notably, the Weil conjectures.

Weil famously suggested that his conjectures might be related to some imaginary cohomology theory, that later became known as ℓ-adic cohomology:

$$|X(k)| = \sum_{i=0}^{2d} (-1)^i \operatorname{tr}\left(\mathsf{Frob}_k \mid \mathrm{H}_c^i(X_{\bar{k}}, \mathbb{Q}_\ell)\right).$$

Climbing the categorical ladder

Using these cohomology groups, Grothendieck proved a large part of Weil's conjectures. Then, Deligne proved a particularly elegant generalization of the remaining one.

He defined a notion of weights for a complex in $\mathrm{D}^b_c(X_{\bar{k}}, \mathbb{Q}_\ell)$ and proved that the functor $\mathrm{R}f_!$ preserves complexes of weights $\leq w$.

Not only this generalizes the remaining conjecture, but it also gives the following useful bound:

$$||X(\mathbb{F}_q)| - q^{nd}| \leq Cq^{n(2d-1)/2}.$$

$$\begin{tikzcd}
\mathrm{D}^b_c(X_{\bar{k}}, \mathbb{Q}_\ell) \arrow[d, rightsquigarrow, "\mathrm{Hom}"] \\
\mathrm{H}^i_c(X_{\bar{k}}, \mathbb{Q}_\ell) \arrow[d, rightsquigarrow, "\mathrm{tr(Frob)}"] \\
|X(k)|
\end{tikzcd}$$

Climbing the categorical ladder

Using these cohomology groups, Grothendieck proved a large part of Weil's conjectures. Then, Deligne proved a particularly elegant generalization of the remaining one.

He defined a notion of weights for a complex in $\mathrm{D}^b_{\mathrm{c}}(X_{\bar{k}}, \mathbb{Q}_\ell)$ and proved that the functor $\mathrm{R}f_!$ preserves complexes of weights $\leq w$.

Not only this generalizes the remaining conjecture, but it also gives the following useful bound:

$$\left||X(\mathbb{F}_q)| - q^{nd}\right| \leq Cq^{n(2d-1)/2}.$$

$$\begin{tikzcd}
\mathrm{D}^b_{\mathrm{c}}(X_{\bar{k}}, \mathbb{Q}_\ell) \arrow[d, rightsquigarrow, "\mathrm{Hom}"] \\
\mathrm{H}^i_c(X_{\bar{k}}, \mathbb{Q}_\ell) \arrow[d, rightsquigarrow, "\mathrm{tr}(\mathrm{Frob})"] \\
|X(k)|
\end{tikzcd}$$

Climbing the categorical ladder

Using these cohomology groups, Grothendieck proved a large part of Weil's conjectures. Then, Deligne proved a particularly elegant generalization of the remaining one.

He defined a notion of weights for a complex in $\mathrm{D}^b_{\mathrm{c}}(X_{\bar{k}}, \mathbb{Q}_\ell)$ and proved that the functor $\mathrm{R}f_!$ preserves complexes of weights $\leq w$.

Not only this generalizes the remaining conjecture, but it also gives the following useful bound:

$$||X(\mathbb{F}_q)| - q^{nd}| \leq Cq^{n(2d-1)/2}.$$

$$\begin{tikzcd}
\mathrm{D}^b_{\mathrm{c}}(X_{\bar{k}}, \mathbb{Q}_\ell) \arrow[d, rightsquigarrow, "\mathrm{Hom}"] \\
\mathrm{H}^i_c(X_{\bar{k}}, \mathbb{Q}_\ell) \arrow[d, rightsquigarrow, "\mathrm{tr}(\mathrm{Frob})"] \\
|X(k)|
\end{tikzcd}$$

Climbing the categorical ladder

Using these cohomology groups, Grothendieck proved a large part of Weil's conjectures. Then, Deligne proved a particularly elegant generalization of the remaining one.

He defined a notion of weights for a complex in $\mathrm{D}^b_{\mathrm{c}}(X_{\bar{k}}, \mathbb{Q}_\ell)$ and proved that the functor $\mathrm{R}f_!$ preserves complexes of weights $\leq w$.

Not only this generalizes the remaining conjecture, but it also gives the following useful bound:

$$||X(\mathbb{F}_q)| - q^{nd}| \leq Cq^{n(2d-1)/2}.$$

$$\begin{tikzcd}
\mathrm{D}^b_{\mathrm{c}}(X_{\bar{k}}, \mathbb{Q}_\ell) \arrow[d, rightsquigarrow, "\mathrm{Hom}"] \\
\mathrm{H}^i_c(X_{\bar{k}}, \mathbb{Q}_\ell) \arrow[d, rightsquigarrow, "\mathrm{tr(Frob)}"] \\
|X(k)|
\end{tikzcd}$$

Correspondences and six-functor formalisms

The *yoga* of six operations

For a long time, six-functor formalisms had no rigorous definition; you knew one when you saw it. Here's what happens in all examples:

- You have a certain (possibly ∞-)category *of spaces* C.
- For every space X in C, there is a closed symmetric monoidal stable ∞-category $D(X)$ (i.e., we have $\otimes$ and $\underline{\mathrm{Hom}}$ functors, which are adjoint).
- For every morphism $f: X \to S$ in C, we have a symmetric monoidal functor $f^*: D(S) \to D(X)$ that admits a right adjoint f_*.
- There exists a collection E of morphisms in C such that, for every morphism $f: X \to S$ in E, we have adjoint functors

$$f_!: D(X) \rightleftarrows D(S): f^!$$

The *yoga* of six operations

For a long time, six-functor formalisms had no rigorous definition; you knew one when you saw it. Here's what happens in all examples:

- You have a certain (possibly ∞-)category *of spaces* C.
- For every space X in C, there is a closed symmetric monoidal stable ∞-category $\mathrm{D}(X)$ (i.e., we have $\otimes$ and $\underline{\mathrm{Hom}}$ functors, which are adjoint).
- For every morphism $f : X \to S$ in C, we have a symmetric monoidal functor $f^* : \mathrm{D}(S) \to \mathrm{D}(X)$ that admits a right adjoint f_*.
- There exists a collection E of morphisms in C such that, for every morphism $f : X \to S$ in E, we have adjoint functors

$$f_! : \mathrm{D}(X) \rightleftarrows \mathrm{D}(S) : f^!.$$

The *yoga* of six operations

For a long time, six-functor formalisms had no rigorous definition; you knew one when you saw it. Here's what happens in all examples:

- You have a certain (possibly ∞-)category *of spaces* C.
- For every space X in C, there is a closed symmetric monoidal stable ∞-category $\mathrm{D}(X)$ (i.e., we have $\otimes$ and $\underline{\mathrm{Hom}}$ functors, which are adjoint).
- For every morphism $f\colon X \to S$ in C, we have a symmetric monoidal functor $f^*\colon \mathrm{D}(S) \to \mathrm{D}(X)$ that admits a right adjoint f_*.
- There exists a collection E of morphisms in C such that, for every morphism $f\colon X \to S$ in E, we have adjoint functors

$$f_!\colon \mathrm{D}(X) \rightleftarrows \mathrm{D}(S) : f^!.$$

The *yoga* of six operations

For a long time, six-functor formalisms had no rigorous definition; you knew one when you saw it. Here's what happens in all examples:

- You have a certain (possibly ∞-)category *of spaces* C.
- For every space X in C, there is a closed symmetric monoidal stable ∞-category $\mathsf{D}(X)$ (i.e., we have $\otimes$ and $\underline{\mathrm{Hom}}$ functors, which are adjoint).
- For every morphism $f: X \to S$ in C, we have a symmetric monoidal functor $f^*: \mathsf{D}(S) \to \mathsf{D}(X)$ that admits a right adjoint f_*.
- There exists a collection E of morphisms in C such that, for every morphism $f: X \to S$ in E, we have adjoint functors

$$f_!: \mathsf{D}(X) \rightleftarrows \mathsf{D}(S) : f^!.$$

The *yoga* of six operations

For a long time, six-functor formalisms had no rigorous definition; you knew one when you saw it. Here's what happens in all examples:

- You have a certain (possibly ∞-)category *of spaces* C.
- For every space X in C, there is a closed symmetric monoidal stable ∞-category $\mathrm{D}(X)$ (i.e., we have $\otimes$ and $\underline{\mathrm{Hom}}$ functors, which are adjoint).
- For every morphism $f\colon X \to S$ in C, we have a symmetric monoidal functor $f^*\colon \mathrm{D}(S) \to \mathrm{D}(X)$ that admits a right adjoint f_*.
- There exists a collection E of morphisms in C such that, for every morphism $f\colon X \to S$ in E, we have adjoint functors

$$f_!\colon \mathrm{D}(X) \rightleftarrows \mathrm{D}(S) : f^!.$$

Base change and the projection formula

These functors should satisfy certain compatibilities. Notably:

- (Projection formula) For f in E, there exists a natural isomorphism
$$f_! - \otimes - \simeq f_!(- \otimes f^* -).$$
- (Base change) Given a cartesian diagram in C
$$\begin{tikzcd} X' \arrow[r, "g'"] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ S' \arrow[r, "g"'] & S, \end{tikzcd}$$
with f and f' in E, we have a natural isomorphism
$$g^* f_! \simeq f'_! g'^{,*}.$$

Base change and the projection formula

These functors should satisfy certain compatibilities. Notably:

- (Projection formula) For f in E, there exists a natural isomorphism
$$f_! - \otimes - \simeq f_!(- \otimes f^* -).$$
- (Base change) Given a cartesian diagram in C
$$\begin{tikzcd} X' \arrow[r, "g'"] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ S' \arrow[r, "g"'] & S, \end{tikzcd}$$
with f and f' in E, we have a natural isomorphism
$$g^* f_! \simeq f'_! g'^{,*}.$$

Proper maps and open immersions

We also expect classes of morphisms $J, P \subset E$ such that, for $f \in E$:

- If Δ_f lies in P, there exists a natural transformation $f_! \to f_*$. It is an isomorphism if $f \in P$.
- If Δ_f lies in J, there exists a natural transformation $f^* \to f^!$. It is an isomorphism if $f \in J$.

In motivic settings one often adds even more axioms, but those are the axioms satisfied by *every* six-functor formalism that I know.

Proper maps and open immersions

We also expect classes of morphisms $J, P \subset E$ such that, for $f \in E$:

- If Δ_f lies in P, there exists a natural transformation $f_! \to f_*$. It is an isomorphism if $f \in P$.
- If Δ_f lies in J, there exists a natural transformation $f^* \to f^!$. It is an isomorphism if $f \in J$.

In motivic settings one often adds even more axioms, but those are the axioms satisfied by *every* six-functor formalism that I know.

Proper maps and open immersions

We also expect classes of morphisms $J, P \subset E$ such that, for $f \in E$:

- If Δ_f lies in P, there exists a natural transformation $f_! \to f_*$. It is an isomorphism if $f \in P$.
- If Δ_f lies in J, there exists a natural transformation $f^* \to f^!$. It is an isomorphism if $f \in J$.

In motivic settings one often adds even more axioms, but those are the axioms satisfied by *every* six-functor formalism that I know.

Proper maps and open immersions

We also expect classes of morphisms $J, P \subset E$ such that, for $f \in E$:

- If Δ_f lies in P, there exists a natural transformation $f_! \to f_*$. It is an isomorphism if $f \in P$.
- If Δ_f lies in J, there exists a natural transformation $f^* \to f^!$. It is an isomorphism if $f \in J$.

In motivic settings one often adds even more axioms, but those are the axioms satisfied by *every* six-functor formalism that I know.

How to define a six-functor formalism? Property vs data

Suppose we want to build a six-functor formalism from scratch: what data do we need to provide?

- The ∞-categories $D(X)$, with their symmetric monoidal structures. (Being stable and admitting a inner Hom are properties.)
- Functors $f^* : D(S) \to D(X)$, as well as their symmetric monoidal structures. (Admitting a right adjoint f_* is a property.)
- Functors $f_! : D(X) \to D(S)$. (Admitting a right adjoint $f^!$ is a property.)
- The projection formula $f_! \otimes - \to f_!(- \otimes f^* -)$.
- The base change isomorphisms $g^* f_! \to f'_! g'^*$.
- The forget-supports maps, which can be encoded as natural transformations $f^* f_! \to \mathrm{id}$ and $f_! f^* \to \mathrm{id}$.

How to define a six-functor formalism? Property vs data

Suppose we want to build a six-functor formalism from scratch: what data do we need to provide?

- The ∞-categories $\mathrm{D}(X)$, with their symmetric monoidal structures. (Being stable and admitting a inner Hom are properties.)
- Functors $f^* \colon \mathrm{D}(S) \to \mathrm{D}(X)$, as well as their symmetric monoidal structures. (Admitting a right adjoint f_* is a property.)
- Functors $f_! \colon \mathrm{D}(X) \to \mathrm{D}(S)$. (Admitting a right adjoint $f^!$ is a property.)
- The projection formula $f_! \otimes - \to f_!(- \otimes f^* -)$.
- The base change isomorphisms $g^* f_! \to f'_! g'^*$.
- The forget-supports maps, which can be encoded as natural transformations $f^* f_! \to \mathrm{id}$ and $f_! f^* \to \mathrm{id}$.

How to define a six-functor formalism? Property vs data

Suppose we want to build a six-functor formalism from scratch: what data do we need to provide?

- The ∞-categories $\mathrm{D}(X)$, with their symmetric monoidal structures. (Being stable and admitting a inner Hom are properties.)
- Functors $f^*\colon \mathrm{D}(S) \to \mathrm{D}(X)$, as well as their symmetric monoidal structures. (Admitting a right adjoint f_* is a property.)
- Functors $f_!\colon \mathrm{D}(X) \to \mathrm{D}(S)$. (Admitting a right adjoint $f^!$ is a property.)
- The projection formula $f_! \otimes - \to f_!(- \otimes f^* -)$.
- The base change isomorphisms $g^* f_! \to f'_! g'^*$.
- The forget-supports maps, which can be encoded as natural transformations $f^* f_! \to \mathrm{id}$ and $f_! f^* \to \mathrm{id}$.

How to define a six-functor formalism? Property vs data

Suppose we want to build a six-functor formalism from scratch: what data do we need to provide?

- The ∞-categories $\mathrm{D}(X)$, with their symmetric monoidal structures. (Being stable and admitting a inner Hom are properties.)
- Functors $f^*\colon \mathrm{D}(S) \to \mathrm{D}(X)$, as well as their symmetric monoidal structures. (Admitting a right adjoint f_* is a property.)
- Functors $f_!\colon \mathrm{D}(X) \to \mathrm{D}(S)$. (Admitting a right adjoint $f^!$ is a property.)
- The projection formula $f_! \otimes - \to f_!(- \otimes f^* -)$.
- The base change isomorphisms $g^* f_! \to f'_! g'^{,*}$.
- The forget-supports maps, which can be encoded as natural transformations $f^* f_! \to \mathrm{id}$ and $f_! f^* \to \mathrm{id}$.

How to define a six-functor formalism? Property vs data

Suppose we want to build a six-functor formalism from scratch: what data do we need to provide?

- The ∞-categories $\mathrm{D}(X)$, with their symmetric monoidal structures. (Being stable and admitting a inner Hom are properties.)
- Functors $f^*\colon \mathrm{D}(S) \to \mathrm{D}(X)$, as well as their symmetric monoidal structures. (Admitting a right adjoint f_* is a property.)
- Functors $f_!\colon \mathrm{D}(X) \to \mathrm{D}(S)$. (Admitting a right adjoint $f^!$ is a property.)
- The projection formula $f_! \otimes - \to f_!(- \otimes f^* -)$.
- The base change isomorphisms $g^* f_! \to f'_! g'^*$.
- The forget-supports maps, which can be encoded as natural transformations $f^* f_! \to \mathrm{id}$ and $f_! f^* \to \mathrm{id}$.

How to define a six-functor formalism? Property vs data

Suppose we want to build a six-functor formalism from scratch: what data do we need to provide?

- The ∞-categories $\mathrm{D}(X)$, with their symmetric monoidal structures. (Being stable and admitting a inner Hom are properties.)
- Functors $f^*\colon \mathrm{D}(S) \to \mathrm{D}(X)$, as well as their symmetric monoidal structures. (Admitting a right adjoint f_* is a property.)
- Functors $f_!\colon \mathrm{D}(X) \to \mathrm{D}(S)$. (Admitting a right adjoint $f^!$ is a property.)
- The projection formula $f_! \otimes - \to f_!(- \otimes f^* -)$.
- The base change isomorphisms $g^* f_! \to f'_! g'^{,*}$.
- The forget-supports maps, which can be encoded as natural transformations $f^* f_! \to \mathrm{id}$ and $f_! f^* \to \mathrm{id}$.

How to define a six-functor formalism? Property vs data

Suppose we want to build a six-functor formalism from scratch: what data do we need to provide?

- The ∞-categories $\mathrm{D}(X)$, with their symmetric monoidal structures. (Being stable and admitting a inner Hom are properties.)
- Functors $f^*\colon \mathrm{D}(S) \to \mathrm{D}(X)$, as well as their symmetric monoidal structures. (Admitting a right adjoint f_* is a property.)
- Functors $f_!\colon \mathrm{D}(X) \to \mathrm{D}(S)$. (Admitting a right adjoint $f^!$ is a property.)
- The projection formula $f_! \otimes - \to f_!(- \otimes f^* -)$.
- The base change isomorphisms $g^* f_! \to f'_! g'^{,*}$.
- The forget-supports maps, which can be encoded as natural transformations $f^* f_! \to \mathrm{id}$ and $f_! f^* \to \mathrm{id}$.

This is not enough!

- If $h = g \circ f$, we should provide natural isomorphisms $h^* \to f^*g^*$ and $h_! \to g_!f_!$.
- These isomorphisms should be compatible with the projection formula:

$$\begin{tikzcd}
h_! - \otimes - \arrow[r] \arrow[d] & h_!(- \otimes h^* -) \arrow[d] \arrow[dr] & \\
g_!f_! - \otimes - \arrow[d] & g_!f_!(- \otimes h^* -) \arrow[d] & h_!(- \otimes f^*g^* -). \arrow[dl] \\
g_!(f_! - \otimes g^* -) \arrow[r] & g_!f_!(- \otimes f^*g^* -) &
\end{tikzcd}$$

This is not enough!

- If $h = g \circ f$, we should provide natural isomorphisms $h^* \to f^*g^*$ and $h_! \to g_!f_!$.
- These isomorphisms should be compatible with the projection formula:

$$\begin{tikzcd}
h_! - \otimes - \arrow[r] \arrow[d] & h_!(- \otimes h^* -) \arrow[d] \arrow[dr, bend left] & \\
g_!f_! - \otimes - \arrow[d] & g_!f_!(- \otimes h^* -) \arrow[d] & h_!(- \otimes f^*g^* -). \arrow[dl, bend left] \\
g_!(f_! - \otimes g^* -) \arrow[r] & g_!f_!(- \otimes f^*g^* -) &
\end{tikzcd}$$

Not even this is enough!

- The composition isomorphisms should be compatible with the base change isomorphisms. (I'll spare you the diagram…)
- The base change isomorphisms should be compatible with the projection formula.
- …

Note that, since we are working with ∞-categories, commuting is not a property that some diagram may satisfy; it is extra data that has to be provided.

This is a truly unfathomable amount of data.

Not even this is enough!

- The composition isomorphisms should be compatible with the base change isomorphisms. (I'll spare you the diagram…)
- The base change isomorphisms should be compatible with the projection formula.
- …

Note that, since we are working with ∞-categories, commuting is not a property that some diagram may satisfy; it is extra data that has to be provided.

This is a truly unfathomable amount of data.

Not even this is enough!

- The composition isomorphisms should be compatible with the base change isomorphisms. (I'll spare you the diagram…)
- The base change isomorphisms should be compatible with the projection formula.
- …

Note that, since we are working with ∞-categories, commuting is not a property that some diagram may satisfy; it is extra data that has to be provided.

This is a truly unfathomable amount of data.

Not even this is enough!

- The composition isomorphisms should be compatible with the base change isomorphisms. (I'll spare you the diagram…)
- The base change isomorphisms should be compatible with the projection formula.
- …

Note that, since we are working with ∞-categories, commuting is not a property that some diagram may satisfy; it is extra data that has to be provided.

This is a truly unfathomable amount of data.

Not even this is enough!

- The composition isomorphisms should be compatible with the base change isomorphisms. (I'll spare you the diagram…)
- The base change isomorphisms should be compatible with the projection formula.
- …

Note that, since we are working with ∞-categories, commuting is not a property that some diagram may satisfy; it is extra data that has to be provided.

This is a truly unfathomable amount of data.

Not even this is enough!

- The composition isomorphisms should be compatible with the base change isomorphisms. (I'll spare you the diagram...)
- The base change isomorphisms should be compatible with the projection formula.
- ...

Note that, since we are working with ∞-categories, commuting is not a property that some diagram may satisfy; it is extra data that has to be provided.

This is a truly unfathomable amount of data.

Lurie's brilliant idea: correspondences

Suppose that C is a 1-category admitting fiber products and that $J, P \subset E$ are stable under composition and base change. We will construct a 2-category $\mathrm{Corr}_{J,P}(\mathrm{C}, E)$ of correspondences.

The objects of $\mathrm{Corr}_{J,P}(\mathrm{C}, E)$ are the same as the objects of C. A morphism $X \to Y$ in $\mathrm{Corr}_{J,P}(\mathrm{C}, E)$ amounts to a diagram

$$\begin{tikzcd} & W \arrow[dl] \arrow[dr] & \\ X & & Y, \end{tikzcd}$$

where $W \to Y$ lies in E.

Lurie's brilliant idea: correspondences

Suppose that C is a 1-category admitting fiber products and that $J, P \subset E$ are stable under composition and base change. We will construct a 2-category $\mathrm{Corr}_{J,P}(\mathrm{C}, E)$ of correspondences.

The objects of $\mathrm{Corr}_{J,P}(\mathrm{C}, E)$ are the same as the objects of C. A morphism $X \to Y$ in $\mathrm{Corr}_{J,P}(\mathrm{C}, E)$ amounts to a diagram

$$\begin{tikzcd} & W \arrow[dl] \arrow[dr] & \\ X & & Y, \end{tikzcd}$$

where $W \to Y$ lies in E.

Lurie's brilliant idea: correspondences

Suppose that C is a 1-category admitting fiber products and that $J, P \subset E$ are stable under composition and base change. We will construct a 2-category $\mathrm{Corr}_{J,P}(\mathrm{C}, E)$ of correspondences.

The objects of $\mathrm{Corr}_{J,P}(\mathrm{C}, E)$ are the same as the objects of C. A morphism $X \to Y$ in $\mathrm{Corr}_{J,P}(\mathrm{C}, E)$ amounts to a diagram

$$\begin{tikzcd} & W \arrow[dl] \arrow[dr] & \\ X & & Y, \end{tikzcd}$$

where $W \to Y$ lies in E.

Correspondences II

To compose $X \leftarrow W \rightarrow Y$ and $Y \leftarrow V \rightarrow Z$, we take a fiber product:

$$\begin{tikzcd}
 & W \times_Y V \arrow[dl] \arrow[dr] & \\
W \arrow[d] \arrow[dr] & & V \arrow[dl] \arrow[d] \\
X & Y & Z.
\end{tikzcd}$$

Finally, a 2-morphism $(X \leftarrow W_1 \rightarrow Y) \rightarrow (X \leftarrow W_2 \rightarrow Y)$ is a diagram

$$\begin{tikzcd}
 & W_1 \arrow[dl] \arrow[dr] & \\
X & Z \arrow[u, "p"'] \arrow[d, "j"'] & Y. \\
 & W_2 \arrow[ul] \arrow[ur] &
\end{tikzcd}$$

Correspondences II

To compose $X \leftarrow W \rightarrow Y$ and $Y \leftarrow V \rightarrow Z$, we take a fiber product:

$$\begin{tikzcd}
 & & W \times_Y V \arrow[dl] \arrow[dr] & & \\
 & W \arrow[dl] \arrow[dr] & & V \arrow[dl] \arrow[dr] & \\
X & & Y & & Z.
\end{tikzcd}$$

Finally, a 2-morphism $(X \leftarrow W_1 \rightarrow Y) \rightarrow (X \leftarrow W_2 \rightarrow Y)$ is a diagram

$$\begin{tikzcd}
 & W_1 \arrow[dl] \arrow[dr] & \\
X & Z \arrow[u, "p"'] \arrow[d, "j"] & Y. \\
 & W_2 \arrow[ul] \arrow[ur] &
\end{tikzcd}$$

The 2-category $\mathrm{Corr}_{J,P}(\mathsf{C}, E)$ has a symmetric monoidal structure, where $X \otimes Y := X \times Y$.

Definition (Lurie, Liu–Zheng)

A *three-functor formalism* is a lax symmetric monoidal functor

$$\mathsf{D} \colon \mathrm{Corr}_{J,P}(\mathsf{C}, E) \to \mathrm{Cat}_{\mathrm{st}}.$$

Here, the adjective *lax* means that instead of having isomorphisms $\mathsf{D}(X) \times \mathsf{D}(Y) \simeq \mathsf{D}(X \times Y)$, we only have a natural transformation

$$\boxtimes \colon \mathsf{D}(-) \times \mathsf{D}(-) \to \mathsf{D}(- \times -).$$

The 2-category $\mathrm{Corr}_{J,P}(\mathrm{C}, E)$ has a symmetric monoidal structure, where $X \otimes Y := X \times Y$.

Definition (Lurie, Liu–Zheng)

A three-functor formalism is a lax symmetric monoidal functor

$$\mathrm{D}\colon \mathrm{Corr}_{J,P}(\mathrm{C}, E) \to \mathrm{Cat}_{\mathrm{st}}.$$

Here, the adjective lax means that instead of having isomorphisms $\mathrm{D}(X) \times \mathrm{D}(Y) \simeq \mathrm{D}(X \times Y)$, we only have a natural transformation

$$\boxtimes\colon \mathrm{D}(-) \times \mathrm{D}(-) \to \mathrm{D}(- \times -).$$

The 2-category $\mathrm{Corr}_{J,P}(\mathrm{C}, E)$ has a symmetric monoidal structure, where $X \otimes Y := X \times Y$.

Definition (Lurie, Liu–Zheng)

A three-functor formalism is a lax symmetric monoidal functor

$$\mathrm{D}\colon \mathrm{Corr}_{J,P}(\mathrm{C}, E) \to \mathrm{Cat}_{\mathrm{st}}.$$

Here, the adjective lax means that instead of having isomorphisms $\mathrm{D}(X) \times \mathrm{D}(Y) \simeq \mathrm{D}(X \times Y)$, we only have a natural transformation

$$\boxtimes\colon \mathrm{D}(-) \times \mathrm{D}(-) \to \mathrm{D}(- \times -).$$

Let's unpack this definition

Given a three-functor formalism $\mathrm{D}\colon \mathrm{Corr}_{J,P}(\mathrm{C}, E) \to \mathrm{Cat}_{\mathrm{st}}$, we define:

- $f^*\colon \mathrm{D}(Y) \to \mathrm{D}(X)$ as the image of $Y \xleftarrow{f} X = X$ by D;
- $f_!\colon \mathrm{D}(X) \to \mathrm{D}(Y)$ as the image of $X = X \xrightarrow{f} Y$ by D;
- $\otimes$ as the composition

$$\mathrm{D}(X) \times \mathrm{D}(X) \xrightarrow{\boxtimes} \mathrm{D}(X \times X) \xrightarrow{\Delta_X^*} \mathrm{D}(X),$$

where $\Delta_X\colon X \to X \times X$ is the diagonal map.

Finally...

Definition (Lurie, Liu–Zheng)

A six-functor formalism is a three-functor formalism in which the functors f^*, $f_!$ and $\otimes$ admit right adjoints.

Let's unpack this definition

Given a three-functor formalism $\mathrm{D}\colon \mathrm{Corr}_{J,P}(\mathrm{C}, E) \to \mathrm{Cat}_{\mathrm{st}}$, we define:

- $f^*\colon \mathrm{D}(Y) \to \mathrm{D}(X)$ as the image of $Y \xleftarrow{f} X = X$ by D;
- $f_!\colon \mathrm{D}(X) \to \mathrm{D}(Y)$ as the image of $X = X \xrightarrow{f} Y$ by D;
- $\otimes$ as the composition

$$\mathrm{D}(X) \times \mathrm{D}(X) \xrightarrow{\boxtimes} \mathrm{D}(X \times X) \xrightarrow{\Delta_X^*} \mathrm{D}(X),$$

where $\Delta_X\colon X \to X \times X$ is the diagonal map.

Finally...

Definition (Lurie, Liu–Zheng)

A six-functor formalism is a three-functor formalism in which the functors f^*, $f_!$ and $\otimes$ admit right adjoints.

Let's unpack this definition

Given a three-functor formalism $\mathrm{D}\colon \mathrm{Corr}_{J,P}(\mathrm{C},E) \to \mathrm{Cat}_{\mathrm{st}}$, we define:

- $f^*\colon \mathrm{D}(Y) \to \mathrm{D}(X)$ as the image of $Y \xleftarrow{f} X = X$ by D;
- $f_!\colon \mathrm{D}(X) \to \mathrm{D}(Y)$ as the image of $X = X \xrightarrow{f} Y$ by D;
- $\otimes$ as the composition

$$\mathrm{D}(X) \times \mathrm{D}(X) \xrightarrow{\boxtimes} \mathrm{D}(X \times X) \xrightarrow{\Delta_X^*} \mathrm{D}(X),$$

where $\Delta_X\colon X \to X \times X$ is the diagonal map.

Finally...

Definition (Lurie, Liu–Zheng)

A six-functor formalism is a three-functor formalism in which the functors f^*, $f_!$ and $\otimes$ admit right adjoints.

Let's unpack this definition

Given a three-functor formalism $\mathrm{D}\colon \mathrm{Corr}_{J,P}(\mathrm{C}, E) \to \mathrm{Cat}_{\mathrm{st}}$, we define:

- $f^*\colon \mathrm{D}(Y) \to \mathrm{D}(X)$ as the image of $Y \xleftarrow{f} X = X$ by D;
- $f_!\colon \mathrm{D}(X) \to \mathrm{D}(Y)$ as the image of $X = X \xrightarrow{f} Y$ by D;
- $\otimes$ as the composition

$$\mathrm{D}(X) \times \mathrm{D}(X) \xrightarrow{\boxtimes} \mathrm{D}(X \times X) \xrightarrow{\Delta_X^*} \mathrm{D}(X),$$

  where $\Delta_X\colon X \to X \times X$ is the diagonal map.

Finally...

Definition (Lurie, Liu–Zheng)

A six-functor formalism is a three-functor formalism in which the functors f^*, $f_!$ and $\otimes$ admit right adjoints.

Let's unpack this definition

Given a three-functor formalism $\mathrm{D}\colon \mathrm{Corr}_{J,P}(\mathrm{C}, E) \to \mathrm{Cat}_{\mathrm{st}}$, we define:

- $f^*\colon \mathrm{D}(Y) \to \mathrm{D}(X)$ as the image of $Y \xleftarrow{f} X = X$ by D;
- $f_!\colon \mathrm{D}(X) \to \mathrm{D}(Y)$ as the image of $X = X \xrightarrow{f} Y$ by D;
- $\otimes$ as the composition

$$\mathrm{D}(X) \times \mathrm{D}(X) \xrightarrow{\boxtimes} \mathrm{D}(X \times X) \xrightarrow{\Delta_X^*} \mathrm{D}(X),$$

  where $\Delta_X\colon X \to X \times X$ is the diagonal map.

Finally...

Definition (Lurie, Liu–Zheng)

A six-functor formalism is a three-functor formalism in which the functors f^*, $f_!$ and $\otimes$ admit right adjoints.

How this encodes the compatibilities

This elegant definition contains all the data previously described.

To give an example, consider a cartesian diagram as in the base change axiom and fit it into a composition of correspondences:

$$\begin{tikzcd}
& & X' \arrow[dl, "f'"'] \arrow[dr, "g'"] & & \\
& S' \arrow[dl, equal] \arrow[dr, "g"] & & X \arrow[dl, "f"'] \arrow[dr, equal] & \\
S' & & S & & X.
\end{tikzcd}$$

The functor D maps this composition to $f^*g_!$.

How this encodes the compatibilities

This elegant definition contains all the data previously described.

To give an example, consider a cartesian diagram as in the base change axiom and fit it into a composition of correspondences:

$$
\begin{tikzcd}
 & & X' \arrow[dl, "f'"'] \arrow[dr, "g'"] & & \\
 & S' \arrow[dl, equal] \arrow[dr, "g"] & & X \arrow[dl, "f"'] \arrow[dr, equal] & \\
S' & & S & & X.
\end{tikzcd}
$$

The functor D maps this composition to $f^*g_!$.

How this encodes the compatibilities

This elegant definition contains all the data previously described.

To give an example, consider a cartesian diagram as in the base change axiom and fit it into a composition of correspondences:

$$\begin{tikzcd} & & X' \arrow[dl, "f'"'] \arrow[dr, "g'"] & & \\ & S' \arrow[dl, equal] \arrow[dr, "g"] & & X \arrow[dl, "f"'] \arrow[dr, equal] & \\ S' & & S & & X. \end{tikzcd}$$

The functor D maps this composition to $f^*g_!$.

Base change

On the other hand, we may write the large roof $S' \xleftarrow{f'} X' \xrightarrow{g'} X$ as the composition

$$
\begin{tikzcd}
 & & X' & & \\
 & X' \arrow[ur, equal] \arrow[dl, "f'"'] \arrow[dr, equal] & & X' \arrow[ul, equal] \arrow[dl, equal] \arrow[dr, "g'"] & \\
S' & & X' & & X.
\end{tikzcd}
$$

Hence, the functor D maps it to $g'_! f'^{,*}$. This encodes the base change isomorphism.

All other compatibilities arise from simple manipulations of correspondences!

Base change

On the other hand, we may write the large roof $S' \xleftarrow{f'} X' \xrightarrow{g'} X$ as the composition

$$\begin{tikzcd}
 & & X' \arrow[dl, equal] \arrow[dr, equal] & & \\
 & X' \arrow[dl, "f'"'] \arrow[dr, equal] & & X' \arrow[dl, equal] \arrow[dr, "g'"] & \\
S' & & X' & & X.
\end{tikzcd}$$

Hence, the functor D maps it to $g'_! f'^{,*}$. This encodes the base change isomorphism.

All other compatibilities arise from simple manipulations of correspondences!

Base change

On the other hand, we may write the large roof $S' \xleftarrow{f'} X' \xrightarrow{g'} X$ as the composition

$$\begin{tikzcd}
 & & X' \arrow[dl, equal] \arrow[dr, equal] & & \\
 & X' \arrow[dl, "f'"'] \arrow[dr, equal] & & X' \arrow[dl, equal] \arrow[dr, "g'"] & \\
S' & & X' & & X.
\end{tikzcd}$$

Hence, the functor D maps it to $g'_! f'^{,*}$. This encodes the base change isomorphism.

All other compatibilities arise from simple manipulations of correspondences!

Base change

On the other hand, we may write the large roof $S' \xleftarrow{f'} X' \xrightarrow{g'} X$ as the composition

$$\begin{tikzcd}
 & & X' \arrow[dl, equal] \arrow[dr, equal] & & \\
 & X' \arrow[dl, "f'"'] \arrow[dr, equal] & & X' \arrow[dl, equal] \arrow[dr, "g'"] & \\
S' & & X' & & X.
\end{tikzcd}$$

Hence, the functor D maps it to $g'_! f'^{,*}$. This encodes the base change isomorphism.

All other compatibilities arise from simple manipulations of correspondences!

Deligne's idea to the rescue

Since a functor $\mathrm{D}\colon \mathrm{Corr}_{J,P}(\mathrm{C}, E) \to \mathrm{Cat}_{\mathrm{st}}$ encodes all the data on a six-functor formalism… constructing it must be impossible, right?

It's often simple to construct a functor $\mathrm{C}^{\mathrm{op}} \to \mathrm{CAlg}(\mathrm{Cat}_{\mathrm{st}})$ encoding the functors f^* and $\otimes$. If every map $f \in E$ factors as $f = p \circ j$, for some $j \in J$ and $p \in P$, we may try to construct $p_!$ and $j_!$ separately.

- Since we expect that $p_! \simeq p_*$, the functor $p_!$ should be right adjoint to p^*.
- Similarly, since we expect $j^* \simeq j^!$, the functor $j_!$ should be left adjoint to j^*.

And existence of adjoints is a property, not data that has to be provided.

Deligne's idea to the rescue

Since a functor $\mathsf{D}\colon \mathsf{Corr}_{J,P}(\mathsf{C}, E) \to \mathsf{Cat}_{\mathrm{st}}$ encodes all the data on a six-functor formalism... constructing it must be impossible, right?

It's often simple to construct a functor $\mathsf{C}^{\mathrm{op}} \to \mathsf{CAlg}(\mathsf{Cat}_{\mathrm{st}})$ encoding the functors f^* and $\otimes$. If every map $f \in E$ factors as $f = p \circ j$, for some $j \in J$ and $p \in P$, we may try to construct $p_!$ and $j_!$ separately.

- Since we expect that $p_! \simeq p_*$, the functor $p_!$ should be right adjoint to p^*.
- Similarly, since we expect $j^* \simeq j^!$, the functor $j_!$ should be left adjoint to j^*.

And existence of adjoints is a property, not data that has to be provided.

Deligne's idea to the rescue

Since a functor $\mathrm{D}\colon \mathrm{Corr}_{J,P}(\mathrm{C}, E) \to \mathrm{Cat}_{\mathrm{st}}$ encodes all the data on a six-functor formalism… constructing it must be impossible, right?

It's often simple to construct a functor $\mathrm{C}^{\mathrm{op}} \to \mathrm{CAlg}(\mathrm{Cat}_{\mathrm{st}})$ encoding the functors f^* and $\otimes$. If every map $f \in E$ factors as $f = p \circ j$, for some $j \in J$ and $p \in P$, we may try to construct $p_!$ and $j_!$ separately.

- Since we expect that $p_! \simeq p_*$, the functor $p_!$ should be right adjoint to p^*.
- Similarly, since we expect $j^* \simeq j^!$, the functor $j_!$ should be left adjoint to j^*.

And existence of adjoints is a property, not data that has to be provided.

Deligne's idea to the rescue

Since a functor $\mathrm{D}\colon \mathrm{Corr}_{J,P}(\mathrm{C}, E) \to \mathrm{Cat}_{\mathrm{st}}$ encodes all the data on a six-functor formalism… constructing it must be impossible, right?

It's often simple to construct a functor $\mathrm{C}^{\mathrm{op}} \to \mathrm{CAlg}(\mathrm{Cat}_{\mathrm{st}})$ encoding the functors f^* and $\otimes$. If every map $f \in E$ factors as $f = p \circ j$, for some $j \in J$ and $p \in P$, we may try to construct $p_!$ and $j_!$ separately.

- Since we expect that $p_! \simeq p_*$, the functor $p_!$ should be right adjoint to p^*.
- Similarly, since we expect $j^* \simeq j^!$, the functor $j_!$ should be left adjoint to j^*.

And existence of adjoints is a property, not data that has to be provided.

Deligne's idea to the rescue

Since a functor $\mathrm{D}\colon \mathrm{Corr}_{J,P}(\mathrm{C}, E) \to \mathrm{Cat}_{\mathrm{st}}$ encodes all the data on a six-functor formalism... constructing it must be impossible, right?

It's often simple to construct a functor $\mathrm{C}^{\mathrm{op}} \to \mathrm{CAlg}(\mathrm{Cat}_{\mathrm{st}})$ encoding the functors f^* and $\otimes$. If every map $f \in E$ factors as $f = p \circ j$, for some $j \in J$ and $p \in P$, we may try to construct $p_!$ and $j_!$ separately.

- Since we expect that $p_! \simeq p_*$, the functor $p_!$ should be right adjoint to p^*.
- Similarly, since we expect $j^* \simeq j^!$, the functor $j_!$ should be left adjoint to j^*.

And existence of adjoints is a property, not data that has to be provided.

Deligne's idea to the rescue

Since a functor $\mathsf{D}\colon \mathrm{Corr}_{J,P}(\mathsf{C}, E) \to \mathrm{Cat}_{\mathrm{st}}$ encodes all the data on a six-functor formalism… constructing it must be impossible, right?

It's often simple to construct a functor $\mathsf{C}^{\mathrm{op}} \to \mathrm{CAlg}(\mathrm{Cat}_{\mathrm{st}})$ encoding the functors f^* and $\otimes$. If every map $f \in E$ factors as $f = p \circ j$, for some $j \in J$ and $p \in P$, we may try to construct $p_!$ and $j_!$ separately.

- Since we expect that $p_! \simeq p_*$, the functor $p_!$ should be right adjoint to p^*.
- Similarly, since we expect $j^* \simeq j^!$, the functor $j_!$ should be left adjoint to j^*.

And existence of adjoints is a property, not data that has to be provided.

Did base change and the projection formula become *properties*?

In general, base change and the projection formula are relations between the functors $f_!$ and f^*, which need not be adjoint in any direction. When they are, the adjunction provides natural maps that can be checked to be isomorphisms.

For example, if $j_!$ is left adjoint to j^*, the unit $\eta\colon \mathrm{id} \to j^*j_!$ induces a map

$$\mathrm{Hom}(M \otimes j^*N, M \otimes j^*N) \to \mathrm{Hom}(M \otimes j^*N, j^*j_!M \otimes j^*N)$$

and

$$\begin{aligned}\mathrm{Hom}(M \otimes j^*N, j^*j_!M \otimes j^*N) &\simeq \mathrm{Hom}(M \otimes j^*N, j^*(j_!M \otimes N)) \\ &\simeq \mathrm{Hom}(j_!(M \otimes j^*N), j_!M \otimes N).\end{aligned}$$

Did base change and the projection formula become *properties*?

In general, base change and the projection formula are relations between the functors $f_!$ and f^*, which need not be adjoint in any direction. When they are, the adjunction provides natural maps that can be checked to be isomorphisms.

For example, if $j_!$ is left adjoint to j^*, the unit $\eta\colon \mathrm{id} \to j^*j_!$ induces a map

$$\mathrm{Hom}(M \otimes j^*N, M \otimes j^*N) \to \mathrm{Hom}(M \otimes j^*N, j^*j_!M \otimes j^*N)$$

and

$$\begin{aligned}\mathrm{Hom}(M \otimes j^*N, j^*j_!M \otimes j^*N) &\simeq \mathrm{Hom}(M \otimes j^*N, j^*(j_!M \otimes N)) \\ &\simeq \mathrm{Hom}(j_!(M \otimes j^*N), j_!M \otimes N).\end{aligned}$$

Did base change and the projection formula become *properties*?

In general, base change and the projection formula are relations between the functors $f_!$ and f^*, which need not be adjoint in any direction. When they are, the adjunction provides natural maps that can be checked to be isomorphisms.

For example, if $j_!$ is left adjoint to j^*, the unit $\eta\colon \mathrm{id} \to j^*j_!$ induces a map

$$\mathrm{Hom}(M \otimes j^*N, M \otimes j^*N) \to \mathrm{Hom}(M \otimes j^*N, j^*j_!M \otimes j^*N)$$

and

$$\begin{aligned}\mathrm{Hom}(M \otimes j^*N, j^*j_!M \otimes j^*N) &\simeq \mathrm{Hom}(M \otimes j^*N, j^*(j_!M \otimes N)) \\ &\simeq \mathrm{Hom}(j_!(M \otimes j^*N), j_!M \otimes N).\end{aligned}$$

The construction theorem

This leads to the following definition.

Definition (Ayoub, Liu–Zheng, Mann)

A three-functor seed is a functor $\mathrm{D} \colon \mathrm{C}^{\mathrm{op}} \to \mathrm{CAlg}(\mathrm{Cat}_{\mathrm{st}})$ such that:

- For $j \in J$, the functor j^* admits a left adjoint $j_!$ satisfying base change and the projection formula.
- For $p \in P$, the functor p^* admits a right adjoint $p_!$ satisfying base change and the projection formula.

Theorem (Ayoub, Cisinski–Déglise, Liu–Zheng, Gaitsgory–Rozenblyum, Mann, Dauser–Kuijper, Cnossen–Lenz–Linskens)

Every three-functor seed extends uniquely to a three-functor formalism. (And similarly for six-functor formalisms.)

The construction theorem

This leads to the following definition.

Definition (Ayoub, Liu–Zheng, Mann)

A three-functor seed is a functor $\mathrm{D} \colon \mathrm{C}^{\mathrm{op}} \to \mathrm{CAlg}(\mathrm{Cat}_{\mathrm{st}})$ such that:

- For $j \in J$, the functor j^* admits a left adjoint $j_!$ satisfying base change and the projection formula.
- For $p \in P$, the functor p^* admits a right adjoint $p_!$ satisfying base change and the projection formula.

Theorem (Ayoub, Cisinski–Déglise, Liu–Zheng, Gaitsgory–Rozenblyum, Mann, Dauser–Kuijper, Cnossen–Lenz–Linskens)

Every three-functor seed extends uniquely to a three-functor formalism. (And similarly for six-functor formalisms.)

Six-functor formalisms on smooth schemes

The Riemann–Hilbert correspondence

Let X be a smooth algebraic variety over $\mathbb{C}$. A theorem of Grothendieck asserts that

$$\mathrm{H}^i_{\mathrm{dR}}(X) := \mathrm{H}^i(X, \Omega^\bullet_X) \text{ is isomorphic to } \mathrm{H}^i_{\mathrm{B}}(X) := \mathrm{H}^i(X_{\mathrm{an}}, \mathbb{C}).$$

Both cohomology theories have natural coefficients: holonomic $\mathcal{D}$-modules and constructible sheaves. Grothendieck's theorem can then be enhanced into an isomorphism of six-functor formalisms

$$\mathrm{D}^b_{\mathrm{h}} \simeq \mathrm{D}^b_{\mathrm{c}}.$$

The Riemann–Hilbert correspondence

Let X be a smooth algebraic variety over $\mathbb{C}$. A theorem of Grothendieck asserts that

$$\mathrm{H}^i_{\mathrm{dR}}(X) := \mathrm{H}^i(X, \Omega^\bullet_X) \text{ is isomorphic to } \mathrm{H}^i_{\mathrm{B}}(X) := \mathrm{H}^i(X_{\mathrm{an}}, \mathbb{C}).$$

Both cohomology theories have natural coefficients: holonomic $\mathcal{D}$-modules and constructible sheaves. Grothendieck's theorem can then be enhanced into an isomorphism of six-functor formalisms

$$\mathsf{D}^b_{\mathrm{h}} \simeq \mathsf{D}^b_{\mathrm{c}}.$$

Let X be a smooth algebraic variety over $\mathbb{C}$. A theorem of Grothendieck asserts that

$$\mathrm{H}^i_{\mathrm{dR}}(X) := \mathrm{H}^i(X, \Omega^\bullet_X) \text{ is isomorphic to } \mathrm{H}^i_{\mathrm{B}}(X) := \mathrm{H}^i(X_{\mathrm{an}}, \mathbb{C}).$$

Both cohomology theories have natural coefficients: holonomic $\mathcal{D}$-modules and constructible sheaves. Grothendieck's theorem can then be enhanced into an isomorphism of six-functor formalisms

$$\mathsf{D}^b_{\mathrm{h}} \simeq \mathsf{D}^b_{\mathrm{c}}.$$

de Rham cohomology on singular varieties

If X is singular, $\mathrm{H}^i(X, \Omega_X^\bullet)$ may not agree with $\mathrm{H}^i(X_{\mathrm{an}}, \mathbb{C})$. Similarly, the ring of differential operators $\mathcal{D}_X$ may fail to be noetherian.

This is a problem even if one only cares about smooth varieties: often, one understands a dense open U of a smooth variety X and hopes to recover information about X from the complement $Z := X \setminus U$. But Z is often singular.

de Rham cohomology on singular varieties

If X is singular, $\mathrm{H}^i(X, \Omega_X^\bullet)$ may not agree with $\mathrm{H}^i(X_{\mathrm{an}}, \mathbb{C})$. Similarly, the ring of differential operators $\mathcal{D}_X$ may fail to be noetherian.

This is a problem even if one only cares about smooth varieties: often, one understands a dense open U of a smooth variety X and hopes to recover information about X from the complement $Z := X \setminus U$. But Z is often singular.

Correspondences on smooth varieties

Another problem: one cannot use the formalism of correspondences with only smooth varieties. Indeed, what should be the composition

$$\begin{tikzcd}
 & \mathbb{A}^1 \arrow[dl, equal] \arrow[dr, "x \mapsto x^2"] & & \mathbb{A}^1 \arrow[dl, "x \mapsto x^3"'] \arrow[dr, equal] & \\
\mathbb{A}^1 & & \mathbb{A}^1 & & \mathbb{A}^1 \text{ ?}
\end{tikzcd}$$

(In particular, some work has to be done to properly define the isomorphism $\mathrm{D}^b_{\mathrm{h}} \simeq \mathrm{D}^b_{\mathrm{c}}$.)

Correspondences on smooth varieties

Another problem: one cannot use the formalism of correspondences with only smooth varieties. Indeed, what should be the composition

$$\begin{tikzcd} & \mathbb{A}^1 \arrow[dl, equal] \arrow[dr, "x \mapsto x^2"'] & & \mathbb{A}^1 \arrow[dl, "x \mapsto x^3"] \arrow[dr, equal] & \\ \mathbb{A}^1 & & \mathbb{A}^1 & & \mathbb{A}^1 \end{tikzcd} \quad ?$$

(In particular, some work has to be done to properly define the isomorphism $\mathrm{D}^b_{\mathrm{h}} \simeq \mathrm{D}^b_{\mathrm{c}}$.)

Another problem: one cannot use the formalism of correspondences with only smooth varieties. Indeed, what should be the composition

$$\begin{tikzcd} & \mathbb{A}^1 \arrow[dl, equal] \arrow[dr, "x\mapsto x^2"] & & \mathbb{A}^1 \arrow[dl, "x\mapsto x^3"'] \arrow[dr, equal] & \\ \mathbb{A}^1 & & \mathbb{A}^1 & & \mathbb{A}^1 \end{tikzcd} ?$$

(In particular, some work has to be done to properly define the isomorphism $\mathsf{D}^b_{\mathrm{h}} \simeq \mathsf{D}^b_{\mathrm{c}}$.)

But *every cloud has a silver lining*...

On the other hand, the following definition is perfectly reasonable.

Definition

Fix a base scheme S. A three-functor seed on Sm/S is a functor $\mathrm{D}\colon (\mathrm{Sm}/S)^{\mathrm{op}} \to \mathrm{CAlg}(\mathrm{Cat}_{\mathrm{st}})$ such that:

- For every open immersion j, the functor j^* admits a left adjoint $j_!$ satisfying base change and the projection formula.
- For every proper map p, the functor p^* admits a right adjoint $p_!$ satisfying base change[1] and the projection formula.

We say that D is local if, for every closed immersion $i\colon Z \hookrightarrow X$ in Sm/S with complement $j\colon U \hookrightarrow X$, we have a Verdier sequence

$$\mathrm{D}(Z) \xrightarrow{i_*} \mathrm{D}(X) \xrightarrow{j^*} \mathrm{D}(U).$$

[1] Whenever it exists!

But *every cloud has a silver lining*...

On the other hand, the following definition is perfectly reasonable.

Definition

Fix a base scheme S. A three-functor seed on Sm/S is a functor $\mathrm{D}\colon (\mathrm{Sm}/S)^{\mathrm{op}} \to \mathrm{CAlg}(\mathrm{Cat}_{\mathrm{st}})$ such that:

- For every open immersion j, the functor j^* admits a left adjoint $j_!$ satisfying base change and the projection formula.
- For every proper map p, the functor p^* admits a right adjoint $p_!$ satisfying base change[1] and the projection formula.

We say that D is local if, for every closed immersion $i\colon Z \hookrightarrow X$ in Sm/S with complement $j\colon U \hookrightarrow X$, we have a Verdier sequence

$$\mathrm{D}(Z) \xrightarrow{i_*} \mathrm{D}(X) \xrightarrow{j^*} \mathrm{D}(U).$$

[1] Whenever it exists!

Extending to singular schemes

The locality property suggests a way of extending local three-functor seeds to singular schemes! For a (possibly singular) quasi-projective scheme X, there exists a closed immersion

$$i \colon X \hookrightarrow \widetilde{X},$$

where $\widetilde{X}$ is smooth (an open subscheme of $\mathbb{P}^n$).

Then, denoting by $j \colon U \hookrightarrow \widetilde{X}$ its complement, we may define

$$\mathrm{D}(X) := \ker(j^* \colon \mathrm{D}(\widetilde{X}) \to \mathrm{D}(U)).$$

Without too much work, one can prove that $\mathrm{D}(X)$ admits a natural symmetric monoidal structure.

Extending to singular schemes

The locality property suggests a way of extending local three-functor seeds to singular schemes! For a (possibly singular) quasi-projective scheme X, there exists a closed immersion

$$i\colon X \hookrightarrow \widetilde{X},$$

where $\widetilde{X}$ is smooth (an open subscheme of $\mathbb{P}^n$).

Then, denoting by $j\colon U \hookrightarrow \widetilde{X}$ its complement, we may define

$$\mathrm{D}(X) := \ker(j^*\colon \mathrm{D}(\widetilde{X}) \to \mathrm{D}(U)).$$

Without too much work, one can prove that $\mathrm{D}(X)$ admits a natural symmetric monoidal structure.

Extending to singular schemes

The locality property suggests a way of extending local three-functor seeds to singular schemes! For a (possibly singular) quasi-projective scheme X, there exists a closed immersion

$$i\colon X \hookrightarrow \widetilde{X},$$

where $\widetilde{X}$ is smooth (an open subscheme of $\mathbb{P}^n$).

Then, denoting by $j\colon U \hookrightarrow \widetilde{X}$ its complement, we may define

$$\mathrm{D}(X) := \ker(j^*\colon \mathrm{D}(\widetilde{X}) \to \mathrm{D}(U)).$$

Without too much work, one can prove that $\mathrm{D}(X)$ admits a natural symmetric monoidal structure.

Extending to singular schemes

The locality property suggests a way of extending local three-functor seeds to singular schemes! For a (possibly singular) quasi-projective scheme X, there exists a closed immersion

$$i\colon X \hookrightarrow \widetilde{X},$$

where $\widetilde{X}$ is smooth (an open subscheme of $\mathbb{P}^n$).

Then, denoting by $j\colon U \hookrightarrow \widetilde{X}$ its complement, we may define

$$\mathrm{D}(X) := \ker(j^*\colon \mathrm{D}(\widetilde{X}) \to \mathrm{D}(U)).$$

Without too much work, one can prove that $\mathrm{D}(X)$ admits a natural symmetric monoidal structure.

Ambient-invariance

It is not clear that this construction is independent of the choice of closed immersion $i\colon X \hookrightarrow \widetilde{X}$. We say that D is ambient-invariant if, for all diagrams

$$\begin{tikzcd}
 & \widetilde{X}_1 \arrow[dd, "f"] \\
X \arrow[ur, hook, "i_1"] \arrow[dr, hook, "i_2"'] & \\
 & \widetilde{X}_2
\end{tikzcd}$$

with complements $j_1\colon U_1 \hookrightarrow \widetilde{X}_1$ and $j_2\colon U_2 \hookrightarrow \widetilde{X}_2$, the natural functor

$$\ker(j_2^*\colon \mathrm{D}(\widetilde{X}_2) \to \mathrm{D}(U_2)) \xrightarrow{f^*} \ker(j_1^*\colon \mathrm{D}(\widetilde{X}_1) \to \mathrm{D}(U_1))$$

is an equivalence.

Ambient-invariance

It is not clear that this construction is independent of the choice of closed immersion $i \colon X \hookrightarrow \widetilde{X}$. We say that D is ambient-invariant if, for all diagrams

$$\begin{tikzcd} & \widetilde{X}_1 \arrow[d, "f"] \\ X \arrow[ur, hook, "i_1"] \arrow[dr, hook, "i_2"'] & \\ & \widetilde{X}_2 \end{tikzcd}$$

with complements $j_1 \colon U_1 \hookrightarrow \widetilde{X}_1$ and $j_2 \colon U_2 \hookrightarrow \widetilde{X}_2$, the natural functor

$$\ker(j_2^* \colon \mathrm{D}(\widetilde{X}_2) \to \mathrm{D}(U_2)) \xrightarrow{f^*} \ker(j_1^* \colon \mathrm{D}(\widetilde{X}_1) \to \mathrm{D}(U_1))$$

is an equivalence.

Ambient-invariance for free

Theorem (Ayoub, Gallauer–R.)

If S is the spectrum of a perfect field, every local three-functor seed is ambient-invariant.

The proof relies on the fact that the smooth loci on an algebraic variety over a perfect field is dense.

Ambient-invariance for free

Theorem (Ayoub, Gallauer–R.)

If S is the spectrum of a perfect field, every local three-functor seed is ambient-invariant.

The proof relies on the fact that the smooth loci on an algebraic variety over a perfect field is dense.

The equivalence

Let D be an ambient-invariant local three-functor seed on Sm/S.
Given a general (qcqs) S-scheme X, one may:

- suppose that X is finitely presented, by a limit argument;
- suppose that X is quasi-projective (or even affine), by descent;

to define $\mathrm{D}(X)$.

Theorem (Gallauer–R.)

This construction extends to a three-functor formalism on Sch/S, and gives an equivalence of $(\infty, 2)$-categories

$$\mathrm{TFS}_{\mathrm{inv,loc}}(\mathrm{Sm}/S; J, P) \simeq \mathrm{TFF}_{\mathrm{cont,loc}}(\mathrm{Sch}/S; E, J, P),$$

where E, J, P have their natural meanings.

Let D be an ambient-invariant local three-functor seed on Sm/S. Given a general (qcqs) S-scheme X, one may:

- suppose that X is finitely presented, by a limit argument;
- suppose that X is quasi-projective (or even affine), by descent;

to define $\mathrm{D}(X)$.

Theorem (Gallauer–R.)

This construction extends to a three-functor formalism on Sch/S, and gives an equivalence of $(\infty, 2)$-categories

$$\mathrm{TFS}_{\mathrm{inv,loc}}(\mathrm{Sm}/S; J, P) \simeq \mathrm{TFF}_{\mathrm{cont,loc}}(\mathrm{Sch}/S; E, J, P),$$

where E, J, P have their natural meanings.

The equivalence

Let D be an ambient-invariant local three-functor seed on Sm/S. Given a general (qcqs) S-scheme X, one may:

- suppose that X is finitely presented, by a limit argument;
- suppose that X is quasi-projective (or even affine), by descent;

to define $\mathrm{D}(X)$.

Theorem (Gallauer–R.)

This construction extends to a three-functor formalism on Sch/S, and gives an equivalence of $(\infty, 2)$-categories

$$\mathrm{TFS}_{\mathrm{inv,loc}}(\mathrm{Sm}/S; J, P) \simeq \mathrm{TFF}_{\mathrm{cont,loc}}(\mathrm{Sch}/S; E, J, P),$$

where E, J, P have their natural meanings.

The equivalence

Let D be an ambient-invariant local three-functor seed on Sm/S. Given a general (qcqs) S-scheme X, one may:

- suppose that X is finitely presented, by a limit argument;
- suppose that X is quasi-projective (or even affine), by descent;

to define D(X).

Theorem (Gallauer–R.)

This construction extends to a three-functor formalism on Sch/S, and gives an equivalence of $(\infty, 2)$-categories

$$\mathsf{TFS}_{\mathrm{inv,loc}}(\mathsf{Sm}/S; J, P) \simeq \mathsf{TFF}_{\mathrm{cont,loc}}(\mathsf{Sch}/S; E, J, P),$$

where E, J, P have their natural meanings.

The equivalence

Let D be an ambient-invariant local three-functor seed on Sm/S. Given a general (qcqs) S-scheme X, one may:

- suppose that X is finitely presented, by a limit argument;
- suppose that X is quasi-projective (or even affine), by descent;

to define $\mathrm{D}(X)$.

Theorem (Gallauer–R.)

This construction extends to a three-functor formalism on Sch/S, and gives an equivalence of $(\infty, 2)$-categories

$$\mathrm{TFS}_{\mathrm{inv,loc}}(\mathrm{Sm}/S; J, P) \simeq \mathrm{TFF}_{\mathrm{cont,loc}}(\mathrm{Sch}/S; E, J, P),$$

where E, J, P have their natural meanings.

Let D be an ambient-invariant local three-functor seed on Sm/S. Given a general (qcqs) S-scheme X, one may:

- suppose that X is finitely presented, by a limit argument;
- suppose that X is quasi-projective (or even affine), by descent;

to define $\mathsf{D}(X)$.

Theorem (Gallauer–R.)

This construction extends to a three-functor formalism on Sch/S, and gives an equivalence of $(\infty, 2)$-categories

$$\mathsf{TFS}_{\mathrm{inv,loc}}(\mathsf{Sm}/S; J, P) \simeq \mathsf{TFF}_{\mathrm{cont,loc}}(\mathsf{Sch}/S; E, J, P),$$

where E, J, P have their natural meanings.

Ideas of the proof: the category Em/S

Let QProj/S be the category of quasi-projective S-schemes, and let Em/S be the following category:

- its objects are closed immersions $X \hookrightarrow \widetilde{X}$ in QProj/S with $\widetilde{X}$ smooth;
- a morphism $(X \hookrightarrow \widetilde{X}) \to (Y \hookrightarrow \widetilde{Y})$ is a commutative square

$$\begin{tikzcd} X \arrow[r, hook] \arrow[d] & \widetilde{X} \arrow[d] \\ Y \arrow[r, hook] & \widetilde{Y}. \end{tikzcd}$$

The category Em/S admits natural functors:

$$\begin{array}{ccccc} \mathrm{Sm}/S & \xrightarrow{\quad d \quad} & \mathrm{Em}/S & \xrightarrow{\quad s \quad} & \mathrm{QProj}/S \\ X & \longmapsto & (X \xrightarrow{\mathrm{id}} X) & & \\ & & (X \hookrightarrow \widetilde{X}) & \longmapsto & X. \end{array}$$

Ideas of the proof: the category Em/S

Let QProj/S be the category of quasi-projective S-schemes, and let Em/S be the following category:

- its objects are closed immersions $X \hookrightarrow \widetilde{X}$ in QProj/S with $\widetilde{X}$ smooth;
- a morphism $(X \hookrightarrow \widetilde{X}) \to (Y \hookrightarrow \widetilde{Y})$ is a commutative square

$$\begin{tikzcd} X \arrow[r, hook] \arrow[d] & \widetilde{X} \arrow[d] \\ Y \arrow[r, hook] & \widetilde{Y}. \end{tikzcd}$$

The category Em/S admits natural functors:

$$\begin{array}{ccccc} \mathrm{Sm}/S & \xrightarrow{d} & \mathrm{Em}/S & \xrightarrow{s} & \mathrm{QProj}/S \\ X & \longmapsto & (X \xrightarrow{\mathrm{id}} X) & & \\ & & (X \hookrightarrow \widetilde{X}) & \longmapsto & X. \end{array}$$

Ideas of the proof: the category Em/S

Let QProj/S be the category of quasi-projective S-schemes, and let Em/S be the following category:

- its objects are closed immersions $X \hookrightarrow \widetilde{X}$ in QProj/S with $\widetilde{X}$ smooth;
- a morphism $(X \hookrightarrow \widetilde{X}) \to (Y \hookrightarrow \widetilde{Y})$ is a commutative square

$$\begin{tikzcd} X \arrow[r, hook] \arrow[d] & \widetilde{X} \arrow[d] \\ Y \arrow[r, hook] & \widetilde{Y}. \end{tikzcd}$$

The category Em/S admits natural functors:

$$\begin{array}{ccccc} \mathrm{Sm}/S & \xrightarrow{d} & \mathrm{Em}/S & \xrightarrow{s} & \mathrm{QProj}/S \\ X & \longmapsto & (X \xrightarrow{\mathrm{id}} X) & & \\ & & (X \hookrightarrow \widetilde{X}) & \longmapsto & X. \end{array}$$

Ideas of the proof: the category Em/S

Let QProj/S be the category of quasi-projective S-schemes, and let Em/S be the following category:

- its objects are closed immersions $X \hookrightarrow \widetilde{X}$ in QProj/S with $\widetilde{X}$ smooth;
- a morphism $(X \hookrightarrow \widetilde{X}) \to (Y \hookrightarrow \widetilde{Y})$ is a commutative square

$$\begin{tikzcd} X \arrow[r, hook] \arrow[d] & \widetilde{X} \arrow[d] \\ Y \arrow[r, hook] & \widetilde{Y}. \end{tikzcd}$$

The category Em/S admits natural functors:

$$\begin{array}{ccccc} \mathrm{Sm}/S & \xrightarrow{d} & \mathrm{Em}/S & \xrightarrow{s} & \mathrm{QProj}/S \\ X & \longmapsto & (X \xrightarrow{\mathrm{id}} X) & & \\ & & (X \hookrightarrow \widetilde{X}) & \longmapsto & X. \end{array}$$

Let D be an ambient-invariant local three-functor seed on Sm/S. To obtain a three-functor seed on Sch/S, we extend D progressively.

$$\begin{tikzcd}
(\mathrm{Sm}/S)^{\mathrm{op}} \arrow[r, "d"] \arrow[d, "\mathrm{D}"'] & (\mathrm{Em}/S)^{\mathrm{op}} \arrow[r, "s"] \arrow[dl, dashed, "(1)"] & (\mathrm{QProj}/S)^{\mathrm{op}} \arrow[r, hook] \arrow[dll, dashed, bend left, "(2)"] & (\mathrm{FP}/S)^{\mathrm{op}} \arrow[r, hook] \arrow[dlll, dashed, bend left, "(3)"] & (\mathrm{Sch}/S)^{\mathrm{op}} \arrow[dllll, dashed, bend left, "(4)"] \\
\mathrm{CAlg}(\mathrm{Cat}_{\mathrm{st}}) & & & &
\end{tikzcd}$$

Step (1) is a non-trivial geometric statement, while step (2) is a non-trivial categorical statement. Steps (3) and (4) are simple dévissages, which I'll ignore.

Having finished these four extensions, the result follows from Cnossen–Lenz–Linskens' theorem.

Ideas of the proof: the big picture

Let D be an ambient-invariant local three-functor seed on Sm/S. To obtain a three-functor seed on Sch/S, we extend D progressively.

$$
\begin{tikzcd}
(\mathrm{Sm}/S)^{\mathrm{op}} \arrow[r, "d"] \arrow[d, "\mathrm{D}"'] & (\mathrm{Em}/S)^{\mathrm{op}} \arrow[r, "s"] \arrow[dl, dashed, "(1)"] & (\mathrm{QProj}/S)^{\mathrm{op}} \arrow[r, hook] \arrow[dll, dashed, bend left, "(2)"] & (\mathrm{FP}/S)^{\mathrm{op}} \arrow[r, hook] \arrow[dlll, dashed, bend left, "(3)"] & (\mathrm{Sch}/S)^{\mathrm{op}} \arrow[dllll, dashed, bend left, "(4)"] \\
\mathrm{CAlg}(\mathrm{Cat}_{\mathrm{st}}) & & & &
\end{tikzcd}
$$

Step (1) is a non-trivial geometric statement, while step (2) is a non-trivial categorical statement. Steps (3) and (4) are simple dévissages, which I'll ignore.

Having finished these four extensions, the result follows from Cnossen–Lenz–Linskens' theorem.

Let D be an ambient-invariant local three-functor seed on Sm/S. To obtain a three-functor seed on Sch/S, we extend D progressively.

$$\begin{tikzcd}
(\mathrm{Sm}/S)^{\mathrm{op}} \arrow[r, "d"] \arrow[d, "\mathrm{D}"'] & (\mathrm{Em}/S)^{\mathrm{op}} \arrow[r, "s"] \arrow[dl, dashed, "(1)"] & (\mathrm{QProj}/S)^{\mathrm{op}} \arrow[r, hook] \arrow[dll, dashed, bend left, "(2)"] & (\mathrm{FP}/S)^{\mathrm{op}} \arrow[r, hook] \arrow[dlll, dashed, bend left, "(3)"] & (\mathrm{Sch}/S)^{\mathrm{op}} \arrow[dllll, dashed, bend left, "(4)"] \\
\mathrm{CAlg}(\mathrm{Cat}_{\mathrm{st}}) & & & &
\end{tikzcd}$$

Step (1) is a non-trivial geometric statement, while step (2) is a non-trivial categorical statement. Steps (3) and (4) are simple dévissages, which I'll ignore.

Having finished these four extensions, the result follows from Cnossen–Lenz–Linskens' theorem.

Ideas of the proof: the geometric step

Extending $\mathrm{D}\colon (\mathrm{Sm}/S)^{\mathrm{op}} \to \mathrm{CAlg}(\mathrm{Cat}_{\mathrm{st}})$ to Em/S is easy: for an object $X \hookrightarrow \widetilde{X}$ of Em/S with complement $j\colon U \hookrightarrow \widetilde{X}$, we define

$$\overline{\mathrm{D}}(X \hookrightarrow \widetilde{X}) := \mathrm{D}(\widetilde{X})/j_!\mathrm{D}(U).$$

With standard tools, one checks that this is indeed a functor $\overline{\mathrm{D}}\colon (\mathrm{Em}/S)^{\mathrm{op}} \to \mathrm{CAlg}(\mathrm{Cat}_{\mathrm{st}})$.

The important part of this step is verifying that $\overline{\mathrm{D}}$ satisfies the axioms of a three-functor seed "on the source".

Ideas of the proof: the geometric step

Extending $\mathrm{D}\colon (\mathrm{Sm}/S)^{\mathrm{op}} \to \mathrm{CAlg}(\mathrm{Cat}_{\mathrm{st}})$ to Em/S is easy: for an object $X \hookrightarrow \widetilde{X}$ of Em/S with complement $j\colon U \hookrightarrow \widetilde{X}$, we define

$$\overline{\mathrm{D}}(X \hookrightarrow \widetilde{X}) := \mathrm{D}(\widetilde{X})/j_!\mathrm{D}(U).$$

With standard tools, one checks that this is indeed a functor $\overline{\mathrm{D}}\colon (\mathrm{Em}/S)^{\mathrm{op}} \to \mathrm{CAlg}(\mathrm{Cat}_{\mathrm{st}})$.

The important part of this step is verifying that $\overline{\mathrm{D}}$ satisfies the axioms of a three-functor seed "on the source".

Ideas of the proof: the geometric step

Extending $\mathsf{D}\colon (\mathsf{Sm}/S)^{\mathrm{op}} \to \mathsf{CAlg}(\mathsf{Cat}_{\mathrm{st}})$ to Em/S is easy: for an object $X \hookrightarrow \widetilde{X}$ of Em/S with complement $j\colon U \hookrightarrow \widetilde{X}$, we define

$$\overline{\mathsf{D}}(X \hookrightarrow \widetilde{X}) := \mathsf{D}(\widetilde{X})/j_!\mathsf{D}(U).$$

With standard tools, one checks that this is indeed a functor $\overline{\mathsf{D}}\colon (\mathsf{Em}/S)^{\mathrm{op}} \to \mathsf{CAlg}(\mathsf{Cat}_{\mathrm{st}})$.

The important part of this step is verifying that $\overline{\mathsf{D}}$ satisfies the axioms of a three-functor seed "on the source".

Ideas of the proof: the geometric step II

For example, let (f, g) be the morphism in Em/S represented by the diagram

$$\begin{tikzcd} X \arrow[r, hook] \arrow[d, "f"'] & \widetilde{X} \arrow[d, "g"] \\ Y \arrow[r, hook] & \widetilde{Y}. \end{tikzcd}$$

If f is an open immersion, we have to show that

$$(f, g)^* : \overline{\mathrm{D}}(Y \hookrightarrow \widetilde{Y}) \to \overline{\mathrm{D}}(X \hookrightarrow \widetilde{X})$$

is naturally isomorphic to $(f, \bar{g})^*$, where $\bar{g}$ is also an open immersion.

Ideas of the proof: the geometric step II

For example, let (f,g) be the morphism in Em/S represented by the diagram

$$\begin{tikzcd} X \arrow[r, hook] \arrow[d, "f"'] & \widetilde{X} \arrow[d, "g"] \\ Y \arrow[r, hook] & \widetilde{Y}. \end{tikzcd}$$

If f is an open immersion, we have to show that

$$(f,g)^* \colon \overline{\mathrm{D}}(Y \hookrightarrow \widetilde{Y}) \to \overline{\mathrm{D}}(X \hookrightarrow \widetilde{X})$$

is naturally isomorphic to $(f,\bar{g})^*$, where $\bar{g}$ is also an open immersion.

Categorical digression

Recall that a functor $F\colon \mathsf{C} \to \mathsf{D}$ is said to be a localization if

$$\mathsf{Fun}(\mathsf{D}, \mathsf{E}) \xrightarrow{-\circ F} \mathsf{Fun}_W(\mathsf{C}, \mathsf{E})$$

is an equivalence for all categories E, where

$$W := \{f \in \mathsf{Mor}(\mathsf{C}) \mid F(f) \text{ is an isomorphism}\}$$

and $\mathsf{Fun}_W(\mathsf{C}, \mathsf{E})$ is the category of functors $G\colon \mathsf{C} \to \mathsf{E}$ such that $G(f)$ is an isomorphism for all $f \in W$.

If F satisfies this universal property with respect to all ∞-categories E, we say that F is an ∞-categorical localization.

This is a stronger property!

Categorical digression

Recall that a functor $F\colon \mathsf{C} \to \mathsf{D}$ is said to be a localization if

$$\mathsf{Fun}(\mathsf{D}, \mathsf{E}) \xrightarrow{-\circ F} \mathsf{Fun}_W(\mathsf{C}, \mathsf{E})$$

is an equivalence for all categories E, where

$$W := \{f \in \mathsf{Mor}(\mathsf{C}) \mid F(f) \text{ is an isomorphism}\}$$

and $\mathsf{Fun}_W(\mathsf{C}, \mathsf{E})$ is the category of functors $G\colon \mathsf{C} \to \mathsf{E}$ such that $G(f)$ is an isomorphism for all $f \in W$.

If F satisfies this universal property with respect to all ∞-categories E, we say that F is an ∞-categorical localization.

This is a stronger property!

Categorical digression

Recall that a functor $F\colon \mathsf{C} \to \mathsf{D}$ is said to be a localization if

$$\mathsf{Fun}(\mathsf{D}, \mathsf{E}) \xrightarrow{-\circ F} \mathsf{Fun}_W(\mathsf{C}, \mathsf{E})$$

is an equivalence for all categories E, where

$$W := \{f \in \mathsf{Mor}(\mathsf{C}) \mid F(f) \text{ is an isomorphism}\}$$

and $\mathsf{Fun}_W(\mathsf{C}, \mathsf{E})$ is the category of functors $G\colon \mathsf{C} \to \mathsf{E}$ such that $G(f)$ is an isomorphism for all $f \in W$.

If F satisfies this universal property with respect to all ∞-categories E, we say that F is an ∞-categorical localization.

This is a stronger property!

Ideas of the proof: the categorical step

Proposition

The functor $s\colon \mathrm{Em}/S \to \mathrm{QProj}/S$ is an ∞-categorical localization.

The proof is based on “Hinich’s Key Lemma”: if $F\colon \mathrm{C} \to \mathrm{D}$ is such that $\mathrm{Fun}([n], F)$ has weakly contractible fibers for all $n \geq 0$, then F is an ∞-categorical localization.

As a result, our functor $\overline{\mathrm{D}}\colon (\mathrm{Em}/S)^{\mathrm{op}} \to \mathrm{CAlg}(\mathrm{Cat}_{\mathrm{st}})$ extends uniquely to QProj/S.

Proposition

The functor $s\colon \mathrm{Em}/S \to \mathrm{QProj}/S$ is an ∞-categorical localization.

The proof is based on "Hinich's Key Lemma": if $F\colon \mathrm{C} \to \mathrm{D}$ is such that $\mathbf{Fun}([n], F)$ has weakly contractible fibers for all $n \geq 0$, then F is an ∞-categorical localization.

As a result, our functor $\overline{\mathrm{D}}\colon (\mathrm{Em}/S)^{\mathrm{op}} \to \mathrm{CAlg}(\mathrm{Cat}_{\mathrm{st}})$ extends uniquely to QProj/S.

Ideas of the proof: the categorical step

Proposition

The functor $s\colon \mathrm{Em}/S \to \mathrm{QProj}/S$ is an ∞-categorical localization.

The proof is based on "Hinich's Key Lemma": if $F\colon \mathrm{C} \to \mathrm{D}$ is such that $\mathbf{Fun}([n], F)$ has weakly contractible fibers for all $n \geq 0$, then F is an ∞-categorical localization.

As a result, our functor $\overline{\mathrm{D}}\colon (\mathrm{Em}/S)^{\mathrm{op}} \to \mathrm{CAlg}(\mathrm{Cat}_{\mathrm{st}})$ extends uniquely to QProj/S.

Questions?